

SelAction: mathematical specification of discrete and overlapping selection

Source-based report for `fortran_mac/`

1 October 2026

Abstract

This report translates the selection-response equations in `fortran_mac/` into portable mathematics. It covers the covariance model, information sources, indices, family-adjusted intensity, Bulmer iteration, one- to three-stage discrete selection, discrete inbreeding, and selection across age classes. Numerical approximations and possible source-level discrepancies are identified explicitly. Internal polynomial approximations for normal probabilities and input/output formatting are outside scope.

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1 Notation and computational structure

The source of record is `seldiscrete.f90`, `selovlp.f90`, `selroutines.f90`, `selinbreeding.f90`, and `seltools.f90` in `fortran_mac/`. Trait indices a, b range from 1 to T , sex g is sire s or dam d , stage j ranges from 1 to 3, and age c begins at 1. All variables are centred within the relevant candidate group. Superscript $\top$ is transpose. A cross-covariance matrix need not be symmetric.

Let $\mathbf{A}$ be the vector of true additive genetic values and $\mathbf{v}$ the economic weights, with zeros outside the breeding goal. The aggregate genotype is

$$\mathbf{H} = \mathbf{v}^\top \mathbf{A}, \quad \sigma_H^2 = \mathbf{v}^\top \mathbf{A} \mathbf{v}, \quad \mathbf{A} = \text{Var}(\mathbf{A}). \quad (1)$$

A trait may enter the index, the goal, both, or neither (source flags `i`, `h`, `b`, `n`). Sire and dam indices can differ. Figure 1 shows the calculation order.

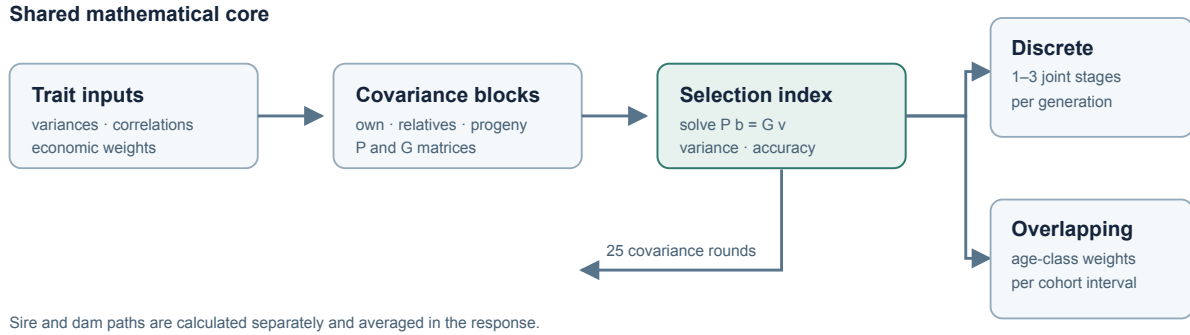


Figure 1: Shared covariance and index calculations feed the discrete and age-structured schemes. Editable XML/SVG source: `selection_flow.svg`.

2 Covariance model and information vector

2.1 Starting components

For `trait a`, the inputs include phenotypic variance $V_{P,a}$, heritability h_a^2 , and common-environment fraction c_a^2 . The code initializes

$$\begin{aligned} V_{A,a} &= h_a^2 V_{P,a}, & V_{C,a} &= c_a^2 V_{P,a}, & V_{E,a} &= V_{P,a} - V_{A,a} - V_{C,a}, \\ V_{A_s,a} &= V_{A_d,a} = \frac{1}{4} V_{A,a}, & V_{A_w,a} &= \frac{1}{2} V_{A,a}. \end{aligned} \quad (2)$$

Given `genetic, phenotypic and common-environment correlation matrices` R_A, R_P, R_C , the initial covariance matrices are

$$\begin{aligned} A_{ab} &= R_{A,ab} \sqrt{V_{A,a} V_{A,b}}, & P_{ab} &= R_{P,ab} \sqrt{V_{P,a} V_{P,b}}, \\ (A_s)_{ab} &= \frac{1}{4} A_{ab}, & (A_d)_{ab} &= \frac{1}{4} A_{ab}, & (A_w)_{ab} &= \frac{1}{2} A_{ab}, \\ C_{ab} &= R_{C,ab} \sqrt{V_{C,a} V_{C,b}}, & E_{ab} &= P_{ab} - A_{ab} - C_{ab}. \end{aligned} \quad (3)$$

For a progeny-test common-environment fraction $c_{\text{prog},a}^2$, the same rule gives

$$C_{\text{prog},ab} = R_{C,ab} \sqrt{c_{\text{prog},a}^2 V_{P,a} c_{\text{prog},b}^2 V_{P,b}}. \quad (4)$$

The source computes an intermediate residual correlation and reconstructs E ; this equals $P - A - C$ when its denominators are nonzero. At every discrete covariance update,

$$\mathbf{A} = \mathbf{A}_s + \mathbf{A}_d + \mathbf{A}_w, \quad \mathbf{P} = \mathbf{A} + \mathbf{C} + \mathbf{E}, \quad \mathbf{F} = \mathbf{A}_s + \mathbf{A}_d + \mathbf{C}, \quad \mathbf{J} = \mathbf{A}_s. \quad (5)$$

F and J are between-full-sib and between-paternal-half-sib phenotype covariances.

2.2 Information sources and the index covariance

Each indexed trait may have **own phenotype** (source 1), **dam EBV** (2), **sire EBV** (3), **up to 20 full-sib means** (4–23), **20 paternal-half-sib means** (24–43), **20 means of dams' EBV in half-sib groups** (44–63), and **20 progeny means** (64–83). The actual observation vector $\mathbf{x}$ consists only of selected trait–source pairs. The routines `info_sources*` create that ordered list; `selection_index` selects its covariance submatrix from an **83-source table** for every trait pair.

Write $\mathbf{P}_x = \text{Var}(\mathbf{x})$ and $\mathbf{G} = \text{Cov}(\mathbf{x}, \mathbf{A})$. For a source measuring trait a and true value A_b , the entries of G are:

Source	$G_{(a,u),b}$
Own phenotype	A_{ab}
Dam EBV	$D_{ba}/2$
Sire EBV	$S_{ba}/2$
Full-sib mean	$(A_s + A_d)_{ab}$
Paternal-half-sib mean	$(A_s)_{ab}$
Mean dam EBV	0
Progeny mean	$A_{ab}/2$

S and D denote the current sire- and dam-EBV covariance matrices used by the implementation. The transpose of their trait indices above follows the `matg` assignments. Goal traits without index information still occupy columns of G .

Family sizes enter $\mathbf{P}_x$ explicitly. For full-sib group u of $n_{F,u}$ offspring, half-sib group u of $n_{H,u}$ offspring from $m_{H,u}$ dams, and a sire's progeny group u of $n_{Q,u}$ offspring from $m_{Q,u}$ dams:

$$\begin{aligned} \text{Var}(\bar{\mathbf{x}}_{F,u}) &= \mathbf{A}_s + \mathbf{A}_d + \mathbf{C} + (\mathbf{A}_w + \mathbf{E})/n_{F,u}, \\ \text{Var}(\bar{\mathbf{x}}_{H,u}) &= \mathbf{A}_s + (\mathbf{A}_d + \mathbf{C})/m_{H,u} + (\mathbf{A}_w + \mathbf{E})/n_{H,u}, \\ \text{Var}(\bar{\mathbf{x}}_{Q,u} \mid \text{sire candidate}) &= \frac{1}{4}\mathbf{A} + (\frac{1}{4}\mathbf{A} + \mathbf{C}_{\text{prog}})/m_{Q,u} + (\mathbf{A}_w + \mathbf{E})/n_{Q,u}, \\ \text{Var}(\bar{\mathbf{x}}_{Q,u} \mid \text{dam candidate}) &= \frac{1}{2}\mathbf{A} + \mathbf{C}_{\text{prog}} + (\mathbf{A}_w + \mathbf{E})/n_{Q,u}^{(d)}. \end{aligned} \quad (6)$$

Distinct full-sib groups have covariance F and distinct half-sib groups have covariance J . Within the same half-sib group, $\text{Cov}(\bar{\mathbf{x}}_{H,u}, \overline{\text{EBV}}_{d,u}) = D/(2m_{H,u})$ and $\text{Var}(\overline{\text{EBV}}_{d,u}) = D/m_{H,u}$. Other cross-blocks, including own phenotype and progeny, are assigned individually in `selection_index`. A port should build that full block table before selecting rows; family means are correlated observations, not independent errors.

3 Shared selection equations

3.1 Index, accuracy and response

The common `selection_index` routine calculates

$$\mathbf{P}_x \mathbf{b} = \mathbf{G} \mathbf{v}, \quad I = \mathbf{b}^\top \mathbf{x}, \quad V_I = \mathbf{b}^\top \mathbf{P}_x \mathbf{b}, \quad r_{IH} = \sqrt{V_I / \sigma_H^2}. \quad (7)$$

A new implementation should solve this system rather than form P_x^{-1} . The source explicitly inverts P_x . With $\mathbf{q} = \mathbf{G}^\top \mathbf{b} = \text{Cov}(\mathbf{A}, I)$ and effective standardized selection differential i_{eff} ,

$$\Delta \mathbf{A} = i_{\text{eff}} \frac{\mathbf{q}}{\sqrt{V_I}}, \quad \Delta H = i_{\text{eff}} \sqrt{V_I}. \quad (8)$$

The second equality follows from the optimal weights. One-stage discrete generation response is the midpoint of the selected sire and dam paths:

$$\Delta \mathbf{A}_{\text{gen}} = \frac{1}{2}(\Delta \mathbf{A}_s + \Delta \mathbf{A}_d), \quad \Delta H_{\text{gen}} = \frac{1}{2}(\Delta H_s + \Delta H_d). \quad (9)$$

3.2 Truncation and related candidates

For upper-tail fraction p , let $t = \Phi^{-1}(1-p)$ and $z = \phi(t)$, with standard-normal CDF Φ and density ϕ . The common `trunc` routine returns

$$i = z/p, \quad k = i(i - t), \quad \text{Var}(Z \mid Z > t) = 1 - k. \quad (10)$$

The discrete one-stage response uses `rawl3` to replace i by a family-adjusted i_{eff} in Eq. (8). Its arguments are $(p, n_w, n_{fs}, n_{hs}, \rho_{fs}, \rho_{hs})$: offspring per full-sib family, full-sib families per paternal-half-sib family, paternal-half-sib families, and two index intraclass correlations. These correlations are quadratic forms

$$\rho_{fs} = \frac{\mathbf{b}^\top \mathbf{R}_{fs} \mathbf{b}}{V_I}, \quad \rho_{hs} = \frac{\mathbf{b}^\top \mathbf{R}_{hs} \mathbf{b}}{V_I}. \quad (11)$$

R_{fs} and R_{hs} are assembled from the same source types as P_x , but for two related candidates. The code clips each correlation to $[-1, 1]$. `rawl3` is an empirical Rawlings/Meuwissen approximation with finite-population branches and interpolation; it is not simply $\phi(t)/p$. Exact numerical reproduction requires porting `rawl3` and `sintvi`, including their family-size branches. Standard-library normal functions can replace `gcef/sabf` if edge conventions are matched.

3.3 Bulmer update

Let $\mathbf{Q} = \mathbf{G}^\top \mathbf{P}_x^{-1} \mathbf{G}$ and $\mathbf{q} = \mathbf{G}^\top \mathbf{b}$. For each sex g , `covariance_update` has the matrix form

$$\mathbf{A}_g^{\text{family, new}} = \frac{1}{4} \left(\mathbf{A} - \frac{k_g}{V_{I,g}} \mathbf{q}_g \mathbf{q}_g^\top \right), \quad (12)$$

$$\mathbf{S}_g^{\text{new}} = \mathbf{Q}_g - \frac{k_g}{V_{I,g}} \mathbf{q}_g \mathbf{u}_g^\top, \quad \mathbf{u}_g = \mathbf{G}_g^\top \mathbf{P}_{x,g}^{-1} \mathbf{G}_g \mathbf{v}.$$

A_g^{family} is A_s or A_d ; S_g is the corresponding sire or dam EBV covariance. In exact arithmetic $u_g = q_g$, but the source evaluates the products separately. After both sexes, Eq. (5) rebuilds A, P, F, J . The within-family additive covariance A_w , common environment C , and residual covariance E remain fixed. The driver executes exactly 25 rounds, with no convergence test: its reported value is the final iterate.

4 Discrete generations

4.1 One stage

`sell1s` builds sire and dam source matrices, initializes Eq. (2), obtains k_s, k_d from Eq. (10), and iterates `selection_index` plus `covariance_update` for 25 rounds. Its initial index variance is a diagonal approximation based on goal-trait inputs; the reported variance comes from Eq. (7) after iteration. Eqs. (8)–(9) give the final trait and goal responses.

4.2 Two and three stages: joint selection

Let $p_{g,j}$ be the fraction passing stage j among candidates entering it and $P_{g,j} = \prod_{\ell=1}^j p_{g,\ell}$ the cumulative fraction. Each stage has an index from Eq. (7), with additional information sources in later stages. The code `runs 25 covariance rounds using the final-stage index`, then reevaluates earlier-stage indices on the resulting matrices. The `final multistage calculation` does not add independent values of Eq. (8).

For `nested indices`, the source approximates standardized index correlations by the accuracy ratio:

$$r_{g,j\ell} = r_{IH,g,j}/r_{IH,g,\ell} \quad (j < \ell), \quad r_{g,j\ell} \leftarrow \min(r_{g,j\ell}, 0.93). \quad (13)$$

The upper cap of 0.93 is source-specific. Let $\mathbf{Z} = (Z_1, \dots, Z_m)$ be standardized stage indices with correlation matrix R_g . Thresholds $t_1, \dots, t_m$ satisfy successive joint-tail probabilities

$$\Pr(Z_1 > t_1) = P_{g,1}, \quad \Pr(Z_1 > t_1, Z_2 > t_2) = P_{g,2}, \quad \Pr(Z_1 > t_1, Z_2 > t_2, Z_3 > t_3) = P_{g,3}. \quad (14)$$

Only the first m equations apply. This is `sseuil1/2/3`; `sdutt1/2/3` evaluate normal tails. Figure 2 shows the intersection of passing sets.

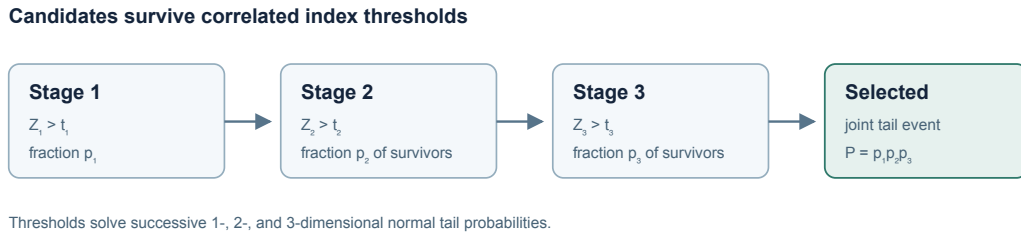


Figure 2: Conditional `survival across stages`; the final selected set is the intersection of all stage-tail events. Editable XML/SVG source: `selection_stages.svg`.

The mathematical target is the conditional index mean vector $\mu_I = E[\mathbf{I} \mid Z_j > t_j \text{ for all } j]$. For two stages, with correlation r , cumulative proportion $P = P_{g,2}$, and $\bar{\Phi} = 1 - \Phi$, define

$$B_1 = \phi(t_1)\bar{\Phi}\left(\frac{t_2 - rt_1}{\sqrt{1-r^2}}\right), \quad B_2 = \phi(t_2)\bar{\Phi}\left(\frac{t_1 - rt_2}{\sqrt{1-r^2}}\right). \quad (15)$$

The implemented means are

$$\mu_{I,1} = \frac{\sqrt{V_{I,1}}}{P}(B_1 + rB_2), \quad \mu_{I,2} = \frac{\sqrt{V_{I,2}}}{P}(rB_1 + B_2). \quad (16)$$

For three stages let $B_j = \phi(t_j) \Pr(Z_{-j} > \mathbf{t}_{-j} \mid Z_j = t_j)$. The two-dimensional conditional tail uses

$$t_{\ell|j} = \frac{t_\ell - r_{j\ell}t_j}{\sqrt{1-r_{j\ell}^2}}, \quad r_{\ell k|j} = \frac{r_{\ell k} - r_{j\ell}r_{jk}}{\sqrt{(1-r_{j\ell}^2)(1-r_{jk}^2)}}. \quad (17)$$

The common boundary-density identity behind the three-stage sums is

$$\mu_{I,j} = \frac{\sqrt{V_{I,j}}}{P_{g,m}} \sum_{\ell=1}^m r_{j\ell} B_\ell, \quad r_{jj} = 1. \quad (18)$$

The three-stage source sets one intermediate conditional correlation, used with stage 2 fixed at its threshold, to zero instead of evaluating Eq. (17). Some B_j terms are recomputed with reversed argument order for later displayed stage contributions. Equation (18) is the coherent mathematical expression; exact emulation should follow the call sequence in `sel3s`.

For trait a , set $\mathbf{c}_a = (\text{Cov}(A_a, I_1), \dots, \text{Cov}(A_a, I_m))^\top$, where $\text{Cov}(A_a, I_j) = \mathbf{b}_j^\top \mathbf{G}_{j, :, a}$. The source uses nested-index covariance $V_{j\ell} = V_{I, \min(j, \ell)}$ and calculates

$$\beta_a = \mathbf{V}^{-1} \mathbf{c}_a, \quad \Delta A_{g,a} = \beta_a^\top \boldsymbol{\mu}_I, \quad \Delta A_{\text{gen},a} = \frac{1}{2}(\Delta A_{s,a} + \Delta A_{d,a}). \quad (19)$$

It also displays Rawlings-adjusted stage results using ratios of `rawl3` to ordinary i at cumulative fractions. These display adjustments are separate from the joint-tail means. Adding individually calculated stage gains double-counts earlier information.

4.3 Discrete-generation inbreeding

The one-stage routine calls `dFmtblup` after its 25 rounds. The overlapping routine does not call it. Let M, F be selected sire and dam counts, $d = F/M$ the mating ratio, and $C_{ss}, C_{sd}, C_{ds}, C_{dd}$ the covariances between offspring information and parental selective advantage constructed in `create_C`. For example, an own-phenotype row of C_{ss} is $\frac{1}{2}A_s^{\text{selected}} + \frac{1}{2d}A_d^{\text{selected}}$; C_{sd} uses $\frac{1}{2}A_s^{\text{selected}} + \frac{1}{2}A_d^{\text{selected}}$. Other source types have their own rows. Define

$$W_s = \mathbf{v}^\top (A_s^{\text{selected}} + A_d^{\text{selected}}/d)\mathbf{v}, \quad W_d = \mathbf{v}^\top (A_s^{\text{selected}} + A_d^{\text{selected}})\mathbf{v}. \quad (20)$$

Two 2×2 matrices are formed from these covariance blocks:

$$\Lambda_{gq} = \frac{1}{2} \frac{i_g \mathbf{b}_g^\top C_{gq} \mathbf{v}}{\sqrt{V_{I,g}} W_q}, \quad \Pi_{gq} = \frac{1}{2} \left(\frac{1}{2} - \frac{k_g \mathbf{b}_g^\top C_{gq} \mathbf{v}}{W_q} \right). \quad (21)$$

With $\mathbf{h} = (1/2, 1/2)^\top$, solve $(I - \Pi^\top) \mathbf{z} = \Lambda^\top \mathbf{h}$, and put $\beta_s = z_s/M$, $\beta_d = z_d/F$. After finite-parent corrections $W'_s = (1 - 1/M)W_s$ and $W'_d = (1 - 1/M)\mathbf{v}^\top A_s^{\text{selected}} \mathbf{v} + (1 - 1/F)\mathbf{v}^\top A_d^{\text{selected}} \mathbf{v}$, the uncorrected result is

$$\Delta F_0 = \frac{1}{2} \left\{ \frac{1}{4M} + M\beta_s^2 W'_s + \frac{1}{4F} + F\beta_d^2 W'_d \right\}. \quad (22)$$

`Poissoncorr` and `hyper_correct` add a finite-family correction. The former replaces each full-/half-sib correlation by $\rho - \rho(1 - \rho^2)(a/M + b/F)$, where $(a, b) = (0.8634, 0.954)$ for full sibs and $(1.4075, 1.4581)$ for half sibs. Below 20 sires it adjusts the selected fraction to $(1 - \rho)p + \rho \max(p, 1/M)$. Each adjusted pair yields a conditional-to-unconditional selection ratio of the form

$$C_{uv} = \frac{\Phi\{(i_u \rho_{uv} - t_v)/\sqrt{1 - k_u \rho_{uv}^2}\}}{p_v}. \quad (23)$$

Six such ratios generate full- and half-sib count covariance matrices in `hyper_correct`. Its additive correction is $\frac{1}{8}[M\delta_s + F\delta_d]$, with $\delta_q = \boldsymbol{\alpha}^\top V_q \boldsymbol{\alpha}$ plus long-term-contribution terms when $M > 19$ and $\boldsymbol{\alpha} = (1/(2M), 1/(2F))^\top$. The six matrix entries and the $M > 19$ branch should be copied from `hyper_correct` for exact numerical agreement. Equation (22) alone is not the displayed final rate.

5 Overlapping generations

5.1 Classes and selection probabilities

`ovlp` stores K sire classes followed by K dam classes. Each has candidate count $N_{g,c}$, selected count $n_{g,c}$, fraction $p_{g,c} = n_{g,c}/N_{g,c}$, index variance $V_{I,g,c}$ and Eq. (10) quantities. With specified counts

$n_{g,c}$ are inputs. Under common-threshold selection, a threshold x_g solves

$$n_{g,c} = N_{g,c} \bar{\Phi} \left(\frac{x_g - m_{g,c}}{\sqrt{V_{I,g,c}}} \right), \quad m_{g,c} = -(c-1) \Delta H_g^{\text{year}}, \quad \sum_c n_{s,c} = M, \quad \sum_c n_{d,c} = F. \quad (24)$$

`trunc_delta` clips each standardized threshold to $[-3, 3]$, and `riddr_root` solves the count equation. A standard survival function and one-dimensional root finder can replace these numerical tools while retaining the clip for numerical agreement.

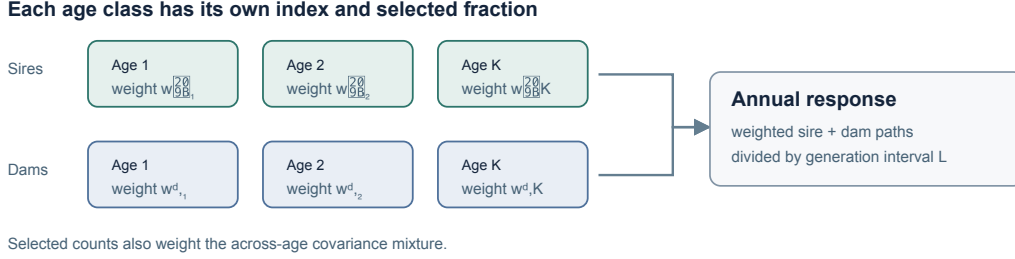


Figure 3: Selected age classes determine each parental path, the generation interval, and the covariance mixture. Editable XML/SVG source: `selection_ages.svg`.

5.2 Generation interval and annual response

Set $N_g^* = \sum_c n_{g,c}$ and $w_{g,c} = n_{g,c}/N_g^*$. The intended generation interval, measured in the source's cohort intervals, is

$$L_s = \sum_c c w_{s,c}, \quad L_d = \sum_c c w_{d,c}, \quad L = (L_s + L_d)/2. \quad (25)$$

For every active class, construct P_x, G, b, V_I with Eq. (7). The annual response is the count-weighted mean of class responses divided by L :

$$\Delta A_a^{\text{year}} = \frac{1}{2L} \sum_{g \in \{s,d\}} \sum_c w_{g,c} i_{g,c} \frac{g_{I,g,c,a}}{\sqrt{V_{I,g,c}}}, \quad \Delta H^{\text{year}} = \frac{1}{2L} \sum_g \sum_c w_{g,c} i_{g,c} \sqrt{V_{I,g,c}}. \quad (26)$$

$g_{I,g,c,a}$ is component a of $G_{g,c}^T b_{g,c}$. Here $i_{g,c}$ is ordinary truncation intensity; overlapping response does not call `raw13`. Thus Eq. (26) reuses Eq. (8) and adds age weights and L . Contrary to the older report, the code does not assemble a gene-flow transition matrix.

5.3 Age-specific covariance update

Each active class receives its own rank-one family covariance update by `ovlp_cov_update`, analogous to Eq. (12). Let $\mu_{g,c,a} = i_{g,c} g_{I,g,c,a} / \sqrt{V_{I,g,c}}$ be the selected trait mean before division by L . The source computes class lag $\ell_{g,c,a} = \mu_{g,c,a} - c \Delta A_{g,a}^{\text{year}}$ and path mean $\bar{\ell}_{g,a} = \sum_c w_{g,c} \ell_{g,c,a}$. Its across-class update is a law-of-total-covariance mixture:

$$(A_g^{\text{family}})_{ab} = \sum_c w_{g,c} \left[(A_{g,c}^{\text{family}})_{ab} + \frac{1}{4} (\ell_{g,c,a} - \bar{\ell}_{g,a})(\ell_{g,c,b} - \bar{\ell}_{g,b}) \right], \quad (27)$$

$$(S_g)_{ab} = \sum_c w_{g,c} \left[(S_{g,c})_{ab} + (\ell_{g,c,a} - \bar{\ell}_{g,a})(\ell_{g,c,b} - \bar{\ell}_{g,b}) \right].$$

It copies mixed matrices into all age classes and repeats 25 times, recalculating selection proportions each round in threshold mode.

Two statements in the current source warrant comparison with fixtures before literal reproduction. The generation-interval loop initializes local accumulators but assigns the sum of a global interval and the current class contribution to them, rather than accumulating across classes. The final per-age P assignment uses local `covapaq`, whose aggregation is not evident in that loop. Equations (25) and (5) specify the coherent mathematical model; a bitwise port needs to test these source statements as implemented.

6 Source map and numerical contract

Source routine	Mathematical responsibility
<code>sel1s, sel2s, sel3s</code>	Discrete input, iteration, and one- to three-stage outputs; Eqs. (9), (14), (19).
<code>selection_index</code>	Source covariance table, P_x, G, b , family correlations and group response; Eqs. (6), (7)–(11).
<code>covariance_update,</code> <code>ovlp_cov_update</code>	Rank-one family and EBV covariance updates; Eqs. (12), (5).
<code>trunc, rawl3</code>	Normal truncation and related-family intensity; Eq. (10) and text below Eq. (11).
<code>sseuil*, sdutt*</code>	Joint-normal thresholds and tails; Eqs. (14)–(18).
<code>ovlp, trunc_delta</code>	Age counts, interval, annual response and mixture; Eqs. (24)–(27).
<code>dFmtblup</code>	Discrete inbreeding; Eqs. (20)–(23).

For a reproducible port, preserve source-row and goal-trait order, separate sire/dam and stage-specific source lists, family-group sizes, selection fractions, 25-round update order, correlation caps, and threshold clipping. Use double precision for joint tails as the source does. Compare index weights and per-trait response against Fortran, not just aggregate gain: a transposed covariance block can conceal itself in the latter.

A Remaining information-source covariance blocks

Here are the remaining P_x rules from `selection_index`. Each entry is a $T \times T$ block; rows corresponding to unused sources are dropped afterward. Write O for own phenotype, B_d, B_s for dam/sire EBV, F_u for full-sib mean, J_u for paternal-half-sib mean, D_u for mean dam EBV in that half-sib group, and Q_u for a progeny mean. Group-diagonal blocks for F_u, J_u, Q_u are in Eq. (6).

Source pair	Covariance block
$O, O; O, B_d; O, B_s$	$P; D/2; S/2$
$B_d, B_d; B_d, B_s; B_s, B_s$	$D; 0; S$
$O, F_u; O, J_u; O, D_u; O, Q_u$	$F; J; 0; A/2$
$B_d, F_u; B_d, J_u; B_d, D_u; B_d, Q_u$	$D/2; 0; 0; D/4$
$B_s, F_u; B_s, J_u; B_s, D_u; B_s, Q_u$	$S/2; S/2; 0; S/4$
$F_u, F_v; F_u, J_v; F_u, D_v; F_u, Q_v$	$F; J; 0; (A_s + A_d)/2$
$J_u, J_v; J_u, D_v; J_u, Q_v$	$J; 0; J/2$
$D_u, D_v; D_u, Q_v$	$0; 0$
Q_u, Q_v	$A/4$ for a sire candidate; $A/2$ for a dam candidate

Here $u \neq v$ where two group indices occur. For the same group, $\text{Var}(D_u) = D/m_{H,u}$ and $\text{Cov}(J_u, D_u) = D/(2m_{H,u})$. These rules include zeros fixed by the source, even where a broader pedigree model could use a nonzero covariance. When switching both trait and source order, transpose the block.

B Family-adjusted intensity approximation

Let $n = n_w n_{fs} n_{hs}$, $n_{\text{sel}} = pn$, and $I_N(p) = \text{shintvi}(p, N)$, the finite-candidate intensity approximation. **raw13** forms three reference intensities: $i_a = I_{\text{nint}(n)}(p)$ for unrelated candidates, i_b for perfectly correlated members within each full-sib family, and i_c for a perfectly correlated paternal half-sib family. For i_b , take family size $h = n_w$ and family count $N = n_{fs} n_{hs}$; for i_c , take $h = n_w n_{fs}$ and $N = n_{hs}$. With $r = \text{mod}(n_{\text{sel}}, h)$ and $a = \text{nint}((n_{\text{sel}} - r)/h) + 1$, the source's reference calculation is

$$I_{\text{family}}(h, N) = \begin{cases} I_{\text{nint}(N)}(1/N), & r = n_{\text{sel}}, \\ \frac{I_{\text{nint}(N)}((a-1)/N) (a-1)(h-r) + I_{\text{nint}(N)}(a/N) ar}{n_{\text{sel}}}, & \text{otherwise.} \end{cases} \quad (28)$$

For real-valued family sizes, the **mod**/**nint** conventions matter. Define

$$f(x) = 0.63e^{3.36(x-1)} + 0.37e^{86(x-1)}. \quad (29)$$

If $n_w = n_{fs} = 1$, $i_{\text{eff}} = i_a$. If $n_{fs} = 1$, the family branch is

$$b_1 = \frac{\log i_c - \log i_a}{\log\{1 - (n_w - 1)/(n - 1)\}}, \quad b = \frac{1}{2}\{1 - f(\rho_{fs})\} + b_1 f(\rho_{fs}), \quad i_{\text{eff}} = i_a \{1 - \rho_{fs}(n_w - 1)/(n - 1)\}^b. \quad (30)$$

If $n_w = 1$, replace n_w by n_{fs} and ρ_{fs} by ρ_{hs} in Eq. (30). If $n_{hs} = 1$, use Eq. (30) with i_b in place of i_c and $\rho_{fs} - \rho_{hs}$ in place of ρ_{fs} , then multiply the result by $\sqrt{1 - \rho_{hs}}$.

For the general branch define

$$\begin{aligned} \rho_a &= \frac{(n_w - 1)\rho_{fs} + (n_{fs} - 1)n_w \rho_{hs}}{n - 1}, & \rho_{bc} &= \frac{(n_w - 1) + (n_{fs} - 1)n_w \rho_{hs}}{n - 1}, \\ \rho_{bc2} &= \frac{\rho_{hs}(n_{fs} - 1)}{n_{fs} n_{hs} - 1}, & \rho_c &= \frac{(n_w - 1) + (n_{fs} - 1)n_w}{n - 1}, & \rho_{c2} &= \frac{n_{fs} - 1}{n_{fs} n_{hs} - 1}. \end{aligned} \quad (31)$$

The interpolation is

$$\begin{aligned}
a_c &= \frac{\log i_c - \log i_a}{\log(1 - \rho_c)}, & b_c &= \frac{\log i_c - \log i_b}{\log(1 - \rho_{c2})}, & y_h &= f(\rho_{hs}), \\
b_0 &= \frac{1}{2}(1 - y_h) + a_c y_h, & i_{bc} &= i_b(1 - \rho_{bc2})^{\frac{1}{2}(1 - y_h) + b_c y_h}, & b_{bc} &= \frac{\log i_{bc} - \log i_a}{\log(1 - \rho_{bc})}, \\
b_a &= \frac{b_0 - b_{bc} y_h}{1 - y_h}, & b &= b_a\{1 - f(\rho_{fs})\} + b_{bc}f(\rho_{fs}), & i_{\text{eff}} &= i_a(1 - \rho_a)^b.
\end{aligned} \tag{32}$$

At $y_h = 1$, the current Mac source sets $b_a = b_{bc}$ to avoid an indeterminate division. `sintvi`'s polynomial and small- p table are numerical approximations, not further population-genetic equations.

C Finite-family inbreeding correction

Denote the six selection ratios from Eq. (23) by full-sib values $C_{ss}^F, C_{sd}^F, C_{dd}^F$ and half-sib values $C_{ss}^J, C_{sd}^J, C_{dd}^J$. Let n_m, n_f be male/female candidates per dam, M, F selected parents, and $T_m = n_m F, T_f = n_f F$. `hyper_correct` forms

$$\begin{aligned}
(V_F)_{11} &= \frac{n_m(n_m - 1)M(M - 1)C_{ss}^F}{T_m(T_m - 1)}, & (V_F)_{12} &= \frac{n_m n_f M F C_{sd}^F}{T_m T_f}, \\
(V_F)_{22} &= \frac{n_f(n_f - 1)F(F - 1)C_{dd}^F}{T_f(T_f - 1)}, & (V_J)_{11} &= \frac{n_m^2 M(M - 1)C_{ss}^J}{T_m(T_m - 1)}, \\
(V_J)_{12} &= \frac{n_m n_f M F C_{sd}^J}{T_m T_f}, & (V_J)_{22} &= \frac{n_f^2 F(F - 1)C_{dd}^J}{T_f(T_f - 1)}.
\end{aligned} \tag{33}$$

Both matrices are symmetric. The source sets $V_s = dV_F + d(d - 1)V_J$ and $V_d = V_F$, then subtracts squared conditional-mean matrices. With the finite-parent W'_s, W'_d from the text and $\lambda_{gq} = \Lambda_{gq}$ from Eq. (21), these matrices are

$$M_s = \begin{pmatrix} 1 + 4\lambda_{ss}^2 W'_s & d(1 + 4\lambda_{ss}\lambda_{ds} W'_s) \\ d(1 + 4\lambda_{ss}\lambda_{ds} W'_s) & d^2(1 + 4\lambda_{ds}^2 W'_s) \end{pmatrix}, \tag{34}$$

$$M_d = \begin{pmatrix} (1 + 4\lambda_{sd}^2 W'_d)/d^2 & (1 + 4\lambda_{sd}\lambda_{dd} W'_d)/d \\ (1 + 4\lambda_{sd}\lambda_{dd} W'_d)/d & 1 + 4\lambda_{dd}^2 W'_d \end{pmatrix}. \tag{35}$$

After subtraction, retain names V_s, V_d and let $\alpha = (1/(2M), 1/(2F))^T$. Set $\delta_q = \alpha^T V_q \alpha$. For $M > 19$, also calculate

$$\delta_s \leftarrow \delta_s + W'_s \mathbf{t}_s^T V_s \mathbf{t}_s, \quad \delta_d \leftarrow \delta_d + W'_d \mathbf{t}_d^T V_d \mathbf{t}_d, \quad \mathbf{t}_q = (\beta_s(2\Pi_{sq}), \beta_d(2\Pi_{dq}))^T. \tag{36}$$

The final predicted rate is

$$\Delta F = \Delta F_0 + \frac{1}{8}(M\delta_s + F\delta_d). \tag{37}$$

The called `dFmtblup` is the copy in `selroutines.f90`; its `trunc` calls use named arguments for intensity and k . A second copy exists in `selinbreeding.f90`.